instruments to fly in the variously colored skies of other worlds. Someday we will go ourselves.

Parachutes have already been used in the atmospheres of Venus and Mars, and are planned for Jupiter and Titan. In 1985 two French-Soviet balloons sailed through the yellow skies of Venus. The *Vega 1* balloon, about 4 meters across, dangled an instrument package 13 meters below. The balloon inflated in the night hemisphere, floated about 54 kilometers above the surface, and transmitted data for almost two Earth days before its batteries failed. In that time it traveled 11,600 kilometers (nearly 7,000 miles) over the surface of Venus, far below. The *Vega 2* balloon had an almost identical profile. The atmosphere of Venus has also been used for aerobraking—changing the *Magellan* spacecraft's orbit by friction with the dense air; this is a key future technology for converting flyby spacecraft to Mars into orbiters and landers.

A Mars mission, scheduled to be launched in 1998, and led by Russia, includes an enormous French hot air balloon—looking something like a vast jellyfish, a Portuguese man-of-war. It's designed to sink to the Martian surface every chilly twilight and rise high when heated by sunlight the next day. The winds are so fast that, if all goes well, it will be carried hundreds of kilometers each day, hopping and skipping over the north pole. In the early morning, when close to the ground, it will obtain very high resolution pictures and other data. The balloon has an instrumental guide-rope, essential for its stability, conceived and designed by a private membership organization based in Pasadena, California, The Planetary Society.

Since the surface pressure on Mars is approximately that at an altitude of 100,000 feet on Earth, we know we can fly airplanes there. The U-2, for example, or the SR-71 *Blackbird* routinely approaches such low pressures. Aircraft with even larger wingspans have been designed for Mars.

The dream of flight and the dream of space travel are twins, conceived by similar visionaries, dependent on allied technologies, and evolving more or less in tandem. As certain practical and